# Code Assignment: Instructions

**Core Philosophy** : At Sahaj, we prioritize design, thinking process, and engineering judgment over mere functionality. We want to see how you reason - especially in a modern, AI-assisted context. We are **not** evaluating GUI or CLI presentation skills.

### 1. Evaluation Criteria & Design Expectations

We look for code that is **readable, maintainable, robust,** and **verified**.

- **Domain Modelling:** Clearly separate behavior from state. Encapsulate state properly and ensure objects reflect the problem domain.
- **Design Principles:** Apply SOLID principles sensibly. Use KISS and YAGNI pragmatically to avoid unnecessary abstractions.
- **Simplicity:** Prefer clear, simple designs over clever ones. Make conscious, explicit trade-offs.
- **Code Quality:** Refactor for clarity. Ensure the intent is obvious to other developers and the code is easy to extend.

### 2. Testing & Confidence

Confidence in correctness is mandatory. Submissions without it are viewed negatively.

- **Unit Tests:** Strongly encouraged wherever reasonable.
- **TDD:** Appreciated but not mandatory.
- **Validation:** If tests are omitted, you **must** explain how you validated the solution.

### 3. Technology Constraints

**No heavy frameworks** (Spring Boot, Django, REST APIs, DBs). Use platform-agnostic languages and standard libraries only.

### 4. AI-Assisted Coding Policy

AI usage is encouraged, but we evaluate your **judgment, validation, and ownership** of the solution. Transparency is mandatory.

**Required Documentation:**

1. **Assumptions (README.md):** Document all assumptions, design decisions, trade-offs, and alternatives considered.
2. **AI Interaction History:** You must provide the full chat history or prompts used (e.g., ai-chat-history.txt).
3. **Reproducible Setup:** If using complex AI workflows (e.g., agents), you must include a **reproducible setup**. This may include:
    - Markdown files with specific commands.
    - Definitions of Skills or Agents used.
    - Plans or orchestration logic required to reproduce the solution.